

Basic Principles of Medicine 2
Module: Endocrinology and Reproduction.

Lecture No: 24

Flipped Classroom:
Nutritional Disorders in Adolescence and
Childhood_ After lecture

Dr. Mohamed Idries
Email: Midries@sgu.edu

St Georges University ©

Copyright

All year 1 courses materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed or published in any form, printed, electronic or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

View only files, such as lecture recordings, are explicitly excluded from download and creating copies of these recordings by students and other users are strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St George's University. Users of this document identifying potential violations of these regulations are asked to bring their concern to the attention of the author.

Required Prereading for flipped classroom

- Preread the posted pdf document: ‘Required Prereading for Nutritional Disorders in Adolescence and Childhood Flipped Classroom’ before the lecture

Lecture 24: Flipped Class: Nutritional Disorders of Adolescence & Childhood	
SOM.MK.III.BPM2.1.ER.2.BCHM.1076	Define and differentiate the following conditions: Nutrition, wasting, cachexia, marasmus, kwashiorkor, stunting.
SOM.MK.III.BPM2.1.ER.2.BCHM.1077	Describe malnutrition in children and its various forms.
SOM.MK.III.BPM2.1.ER.2.BCHM.1078	Describe anorexia nervosa.
SOM.MK.III.BPM2.1.ER.2.BCHM.1079	Describe childhood overnutrition including risk factors, methods of assessment, basis of management and complications
SOM.MK.III.BPM2.1.ER.2.BCHM.1080	Define undernutrition with respect to macronutrients and micronutrients.
SOM.MK.III.BPM2.1.ER.2.BCHM.1081	Discuss the spectrum of protein-energy malnutrition including clinical features, lab findings and anthropometric measurements. Compare and contrast the following childhood nutritional disorders: Kwashiorkor, Marasmus and Marasmic-kwashiorkor.
SOM.MK.III.BPM2.1.ER.2.BCHM.1082	Explain associated micronutrient deficiency with special reference to calcium, phosphate, magnesium, iodine and selenium.
SOM.MK.III.BPM2.1.ER.2.BCHM.1083	Highlight the complications that may occur in the management of children with undernutrition and the basis for nutritional supplementation.

Objectives

SOM.MK.III.BPM2.1.ER.2.BCHM.1076	Define and differentiate the following conditions: Nutrition, wasting, cachexia, marasmus, kwashiorkor, stunting.
SOM.MK.III.BPM2.1.ER.2.BCHM.1077	Describe malnutrition in children and its various forms.
SOM.MK.III.BPM2.1.ER.2.BCHM.1078	Describe anorexia nervosa.
SOM.MK.III.BPM2.1.ER.2.BCHM.1079	Describe childhood overnutrition including risk factors, methods of assessment, basis of management and complications
SOM.MK.III.BPM2.1.ER.2.BCHM.1080	Define undernutrition with respect to macronutrients and micronutrients.
SOM.MK.III.BPM2.1.ER.2.BCHM.1081	Discuss the spectrum of protein-energy malnutrition including clinical features, lab findings and anthropometric measurements. Compare and contrast the following childhood nutritional disorders: Kwashiorkor, Marasmus and Marasmic-kwashiorkor.
SOM.MK.III.BPM2.1.ER.2.BCHM.1082	Explain associated micronutrient deficiency with special reference to calcium, phosphate, magnesium, iodine and selenium.
SOM.MK.III.BPM2.1.ER.2.BCHM.1083	Highlight the complications that may occur in the management of children with undernutrition and the basis for nutritional supplementation.

Nutrition

- The intake of food necessary for the body's dietary needs especially in growing children and adolescents.



- **Good health** is a combination of a well-balanced diet and regular physical activity.

[27. Nutrition: Overview and Macronutrients | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

Energy and nutrient requirements for children vary depending upon age, sex, and activity level

Age	Energy (kcal/day)	Protein (g/day)	Total fat (g/day)
1 to 3 years			
Boys	850 to 1300	13	30 to 40
Girls	750 to 1250	13	30 to 40
4 to 8 years			
Boys	1400 to 1700	19	25 to 35
Girls	1300 to 1600	19	25 to 35
9 to 13 years			
Boys	1800 to 2300	34	25 to 35
Girls	1700 to 2000	34	25 to 35
14 to 18 years			
Boys	2500 to 2800	52	25 to 35
Girls	2000	46	25 to 35

Adapted from:

1. *The Dietary Reference Intakes, National Academy of Sciences, Washington, DC, 2002.*

Clicker Question

A 79-year-old man comes to the physician for regular assessment. Physical examination shows marked loss of muscle mass in all four extremities. DEXA scan shows a decrease in muscle mass two standard deviations below the mean for young healthy adults. This can be attributed to which of the following physiologic changes?

- ✓ A. Decreased insulin-like growth factor level 20%
- B. Decreased free radical production 20%
- C. Elevated estrogen level 20%
- D. Elevated fat metabolism 20%
- E. Elevated testosterone level 20%

Effect of Nutrition in children

- Strong **hair** : Protein, Vitamin D and Omega -3 fatty acids
- Beneficial for **vision**: Omega- 3 fatty acids, Lutein, zinc and vitamin C and E
- Maintain healthy **muscle** mass: Protein, cholesterol, zinc, B vitamins and iron
- Protect against **free radicals**: Antioxidants such as beta- carotene and vitamins C, E and A
- Protects the **skin** from the effects of toxins: Omega- 3 fatty acids
- Build and maintain strong **bones** and **teeth**: Calcium
- Beneficial for the **digestive** system: Fiber, fruits and vegetables

Nutritional Challenges in childhood

Feeding development and behavioral:

- Self-feeding : toddlers play with food
- Picky eaters : resistant to trying new foods
- Food Restriction because of body image in adolescence
- Overindulgence in Fast food

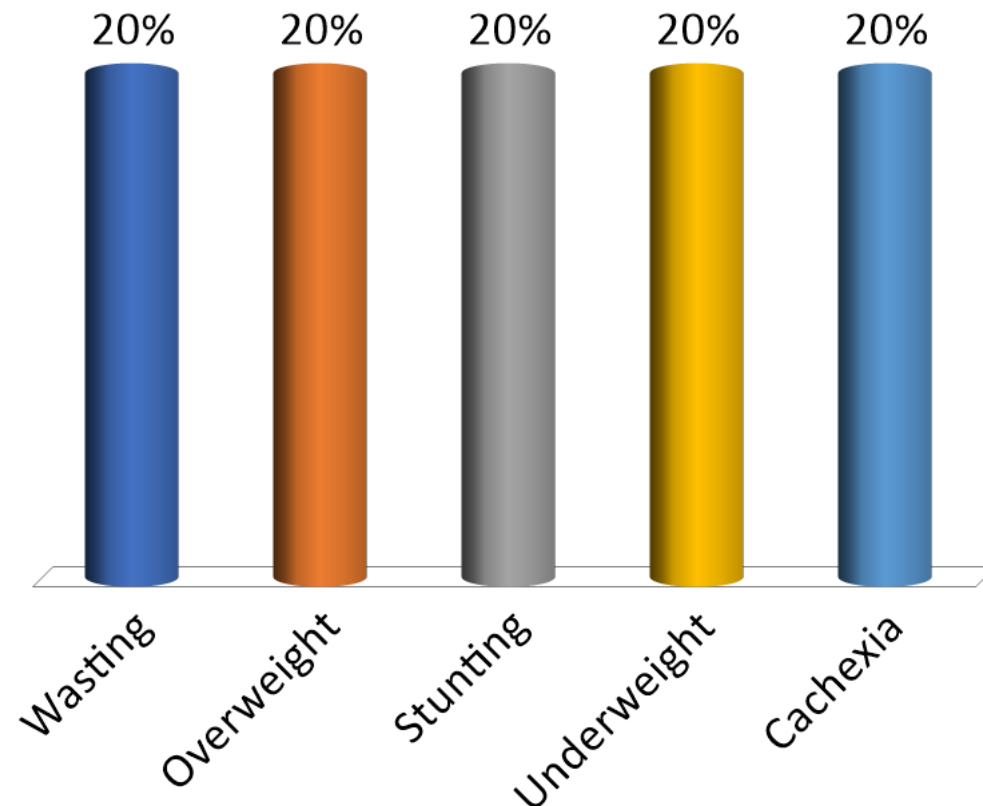
Socio-economical and pathological(Secondary)

- Eating environment: Meals without distractions
- Low economic status or orphaned/ homeless children
- Diabetes, Cystic Fibrosis or Malabsorption Syndrome

Clicker Question

A 2-year-old female child is brought to the clinic by her mother for a routine check-up. She was delivered at 29 weeks and required NICU admission. She has a medical history of Respiratory Distress Syndrome, is up to date with her vaccinations and has a diet history of breast milk only. The physician takes her weight and height measurements, and it is found that her weight-for-age is within normal ranges and her length-for-age is three standard deviations below the normal. Which of the following best describes her condition?

- A. Wasting
- B. Overweight
- ✓ C. Stunting
- D. Underweight
- E. Cachexia



Malnutrition may result from

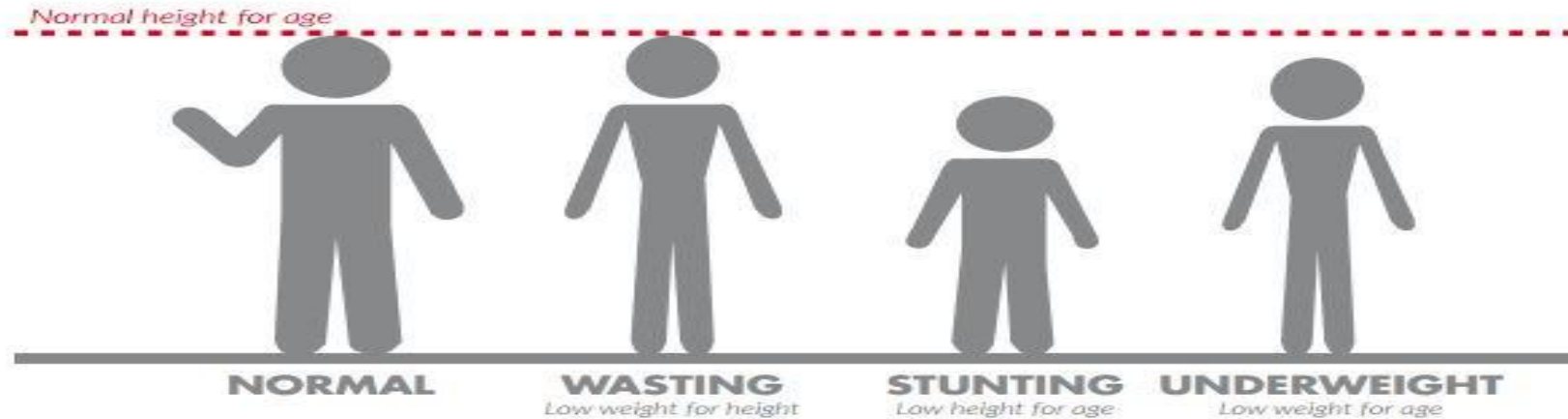
Undernutrition

- **Insufficient intake** of food or of certain nutrients
- inability of the body to **absorb** and use nutrients

Overnutrition

Overconsumption of certain foods.

Abnormal growth and development



WASTING

The acute loss of weight, or failure to gain weight at the expected rate, resulting in a condition of **reduced weight for height**.

STUNTING

The child is short for his age

Commonly seen in chronic malnutrition.

CACHEXIA:

is a hypercatabolic state and a chronic inflammatory response which leads to illness-induced loss of body weight in an individual who has access to food and is not trying to loose weight.

Causes:

- Cancer
- Infection
- Immunodeficiency
- COPD
- Rheumatoid Arthritis

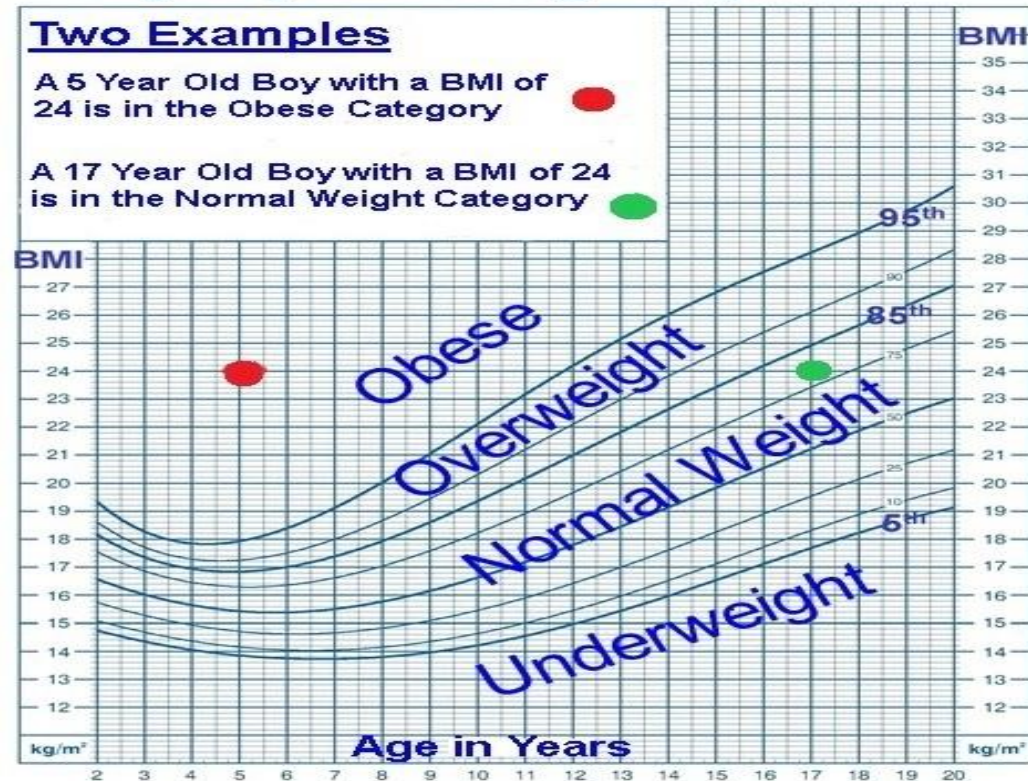
In contrast to simple starvation, which is characterized by a caloric deficiency that can be reversed with appropriate feeding, **cachexia is not reversed by the supplementation of calories.**

BMI Percentile Chart for Boys Age 2 years to Age 20 years

Two Examples

A 5 Year Old Boy with a BMI of 24 is in the Obese Category

A 17 Year Old Boy with a BMI of 24 is in the Normal Weight Category



Published May 30, 2000 (modified 10/16/00).
SOURCE: Developed by the National Center for Health Statistics in collaboration with
the National Center for Chronic Disease Prevention and Health Promotion (2000).
<http://www.cdc.gov/growthcharts>



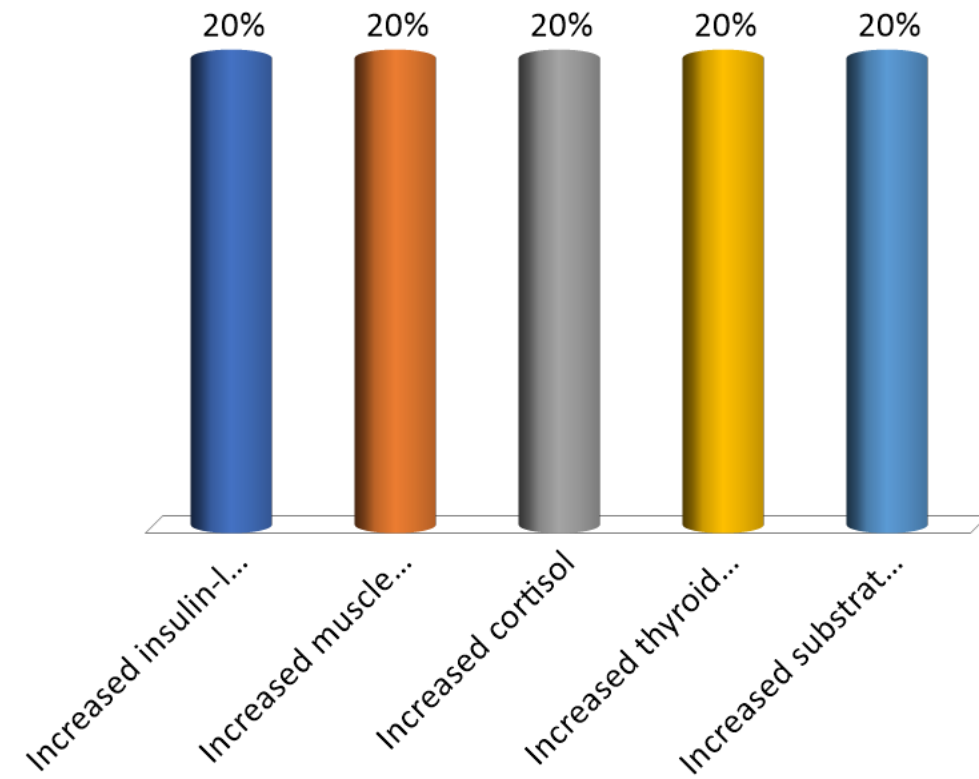
SAFER • HEALTHIER • PEOPLE™

- A child with BMI above the 95th percentile is considered obese
- A child with a BMI between the 85th and 95th percentiles is overweight

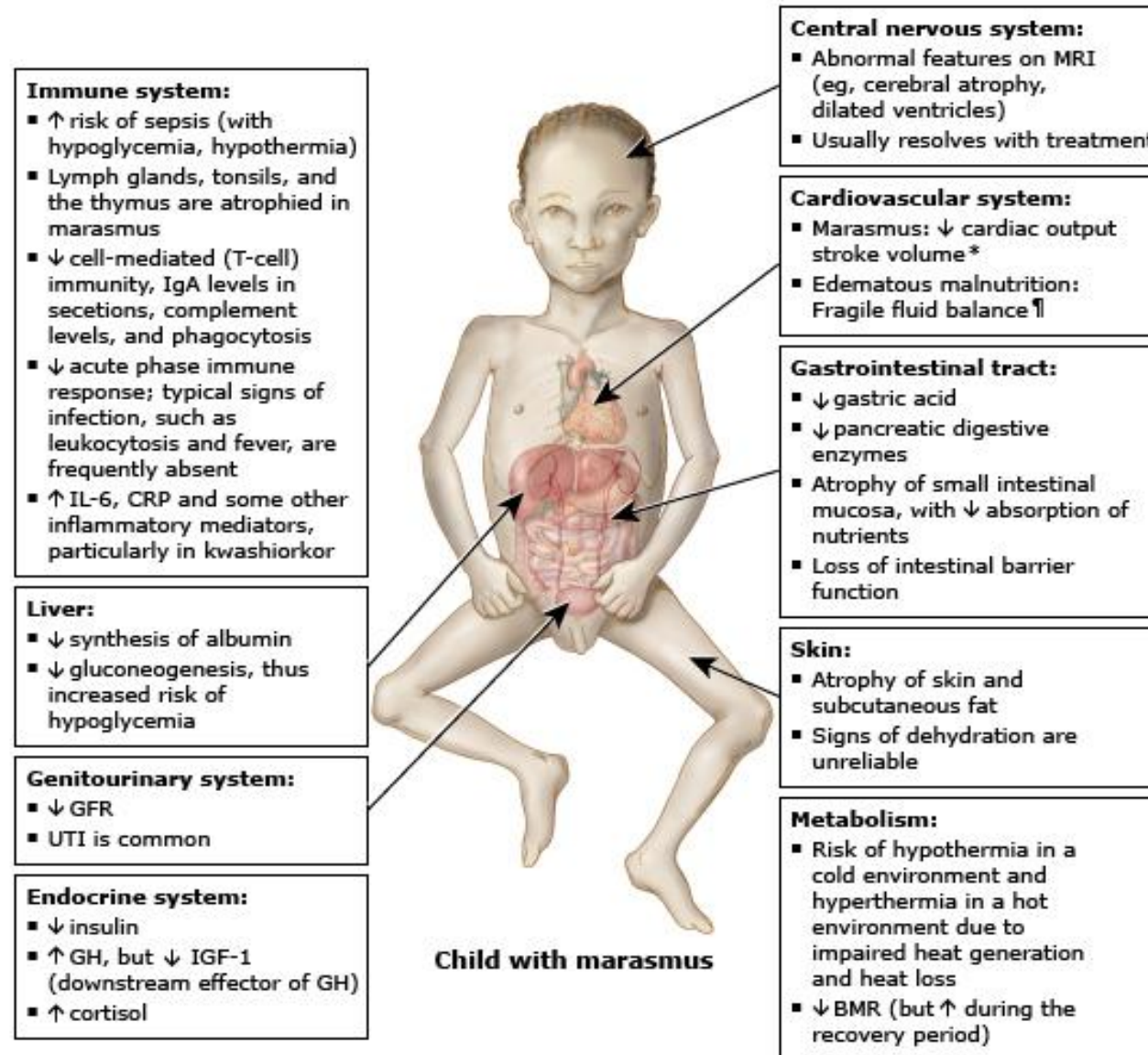
Clicker Question

A 4-year-old male child is brought to the clinic by social services for a 2-month history of neglect. A history is not available at the time. Routine measurements reveal both significantly decreased weight-for-age and height-for-age. Physical examination shows a child who appears wasted and fretful, additionally the child's skin is thin and dry with multiple folds and extremities emaciated. Which of the following pathological processes explains this child's symptoms?

- A. Increased insulin-like growth hormone
- B. Increased muscle glycogen synthesis
- ✓ C. Increased cortisol
- D. Increased thyroid hormone
- E. Increased substrates for glycolysis



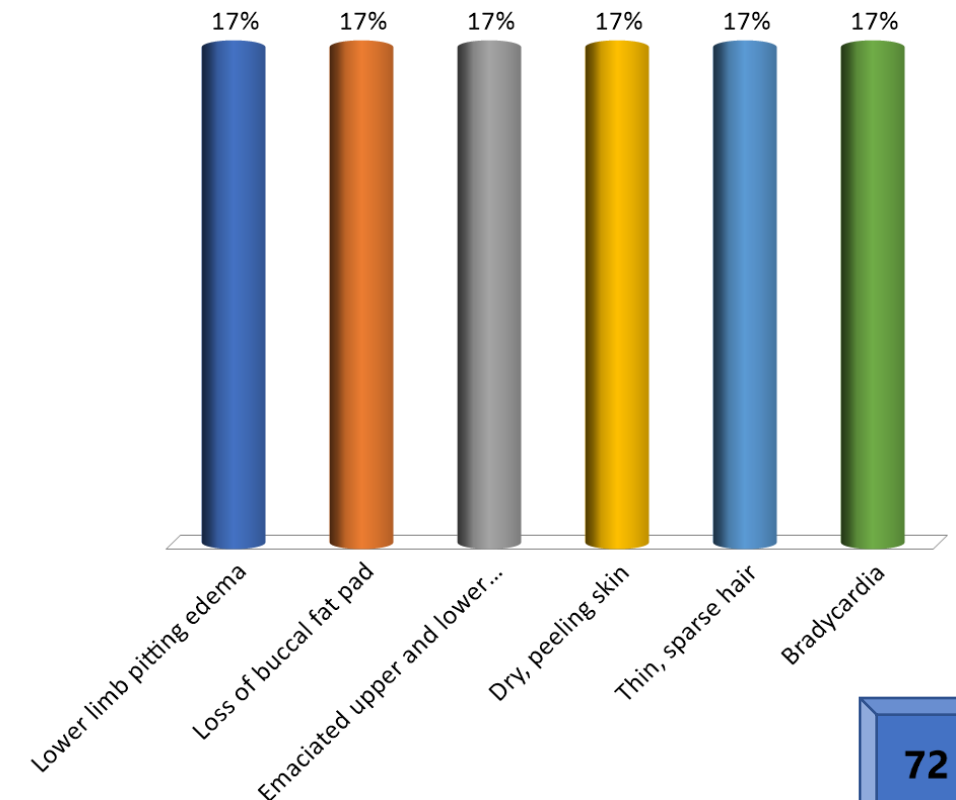
Clinical effects of malnutrition



Clicker Question

A 3-year-old son of immigrant parents is brought by his parents for assessment. Routine measurements reveal both significantly decreased weight-for-height and height-for-age. On physical examination, the head appears large relatively to the rest of the body. Acute severe malnutrition is suspected by the pediatrician. Which of the following is most likely to be absent on clinical assessment of this patient?

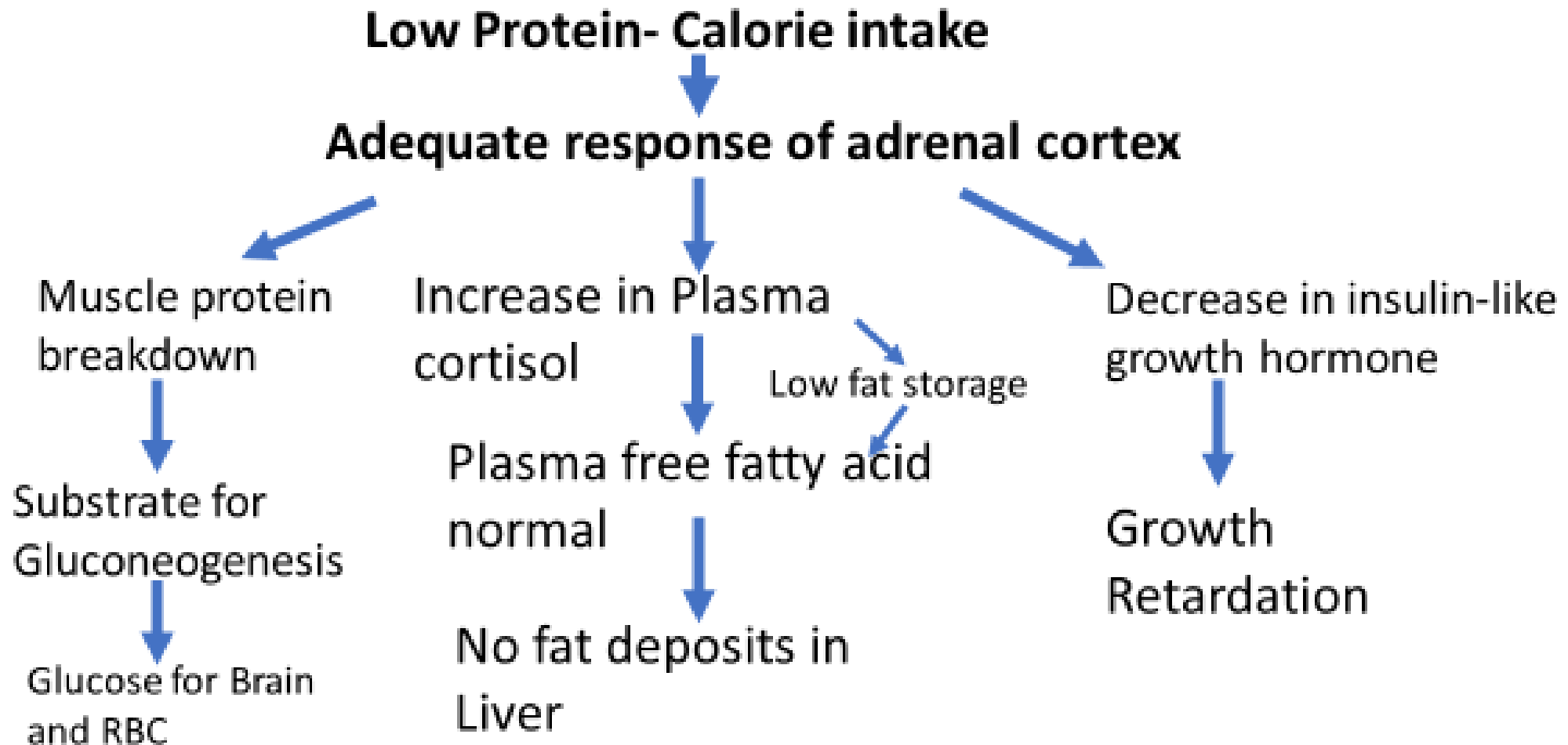
- ✓ A. Lower limb pitting edema
- B. Loss of buccal fat pad
- C. Emaciated upper and lower extremities
- D. Dry, peeling skin
- E. Thin, sparse hair
- F. Bradycardia



MARASMUS (Wasting syndrome)

- Low weight for height
- Wasting of muscle mass leading to decrease mid-upper arm circumference (MUAC)
- Depletion of body fat stores
- Malnutrition without edema
- Gradual onset (months to years)
- Low Calorie intake

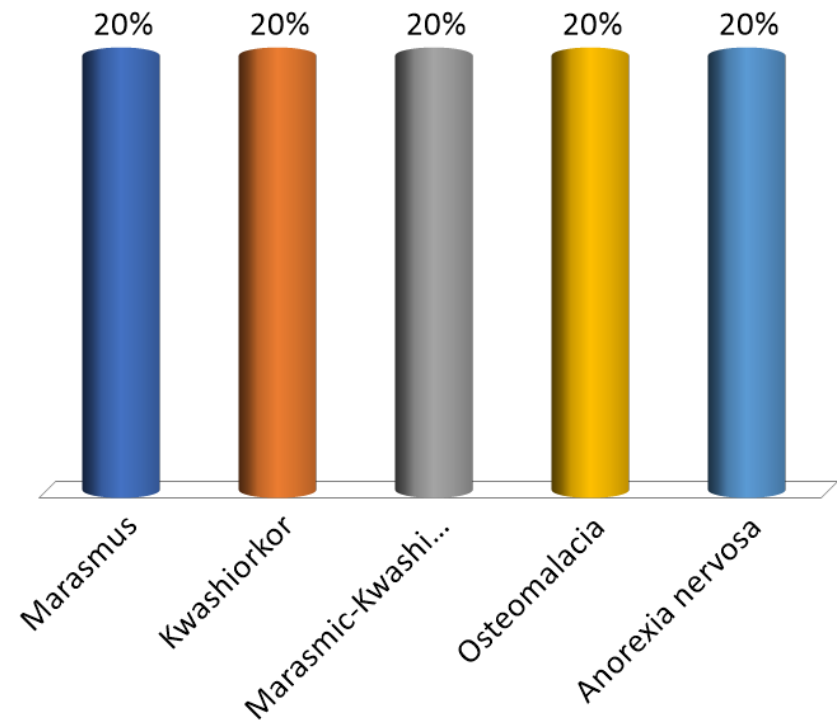
Marasmus



Clicker Question

A 5-year-old male child is brought to the outpatient clinic by his parents. He has a history of two previous admissions for unexplained weakness for which he was discharged with multivitamins and a diet plan. He has no significant birth history, and his vaccinations are up to date. His diet consists mainly of starches and vegetables as the family are vegan. On physical examination, his vitals are normal, his height-for-age is low and he appears uninterested in his surroundings. Further examination reveals that the skin appears hyperpigmentation while the hair is hypopigmented; he also has pitting edema in his extremities with a mildly distended abdomen. What is the most appropriate diagnosis?

- A. Marasmus
- ✓ B. Kwashiorkor
- C. Marasmic-Kwashiorkor
- D. Osteomalacia
- E. Anorexia nervosa



KWASHIOKOR (Edematous Malnutrition)

- Significant protein deprivation despite of adequate energy intake
- Affects **mainly children**
- **Symmetrical Peripheral Pitting Edema**
- Thin, dry, peeling skin with areas of hyperkeratosis and hyperpigmentation
- Dry, dull, hypopigmented hair which often leads to hair loss.
- Distended abdomen with dilated bowels
- Bradycardia, Hypotension and Hypothermia



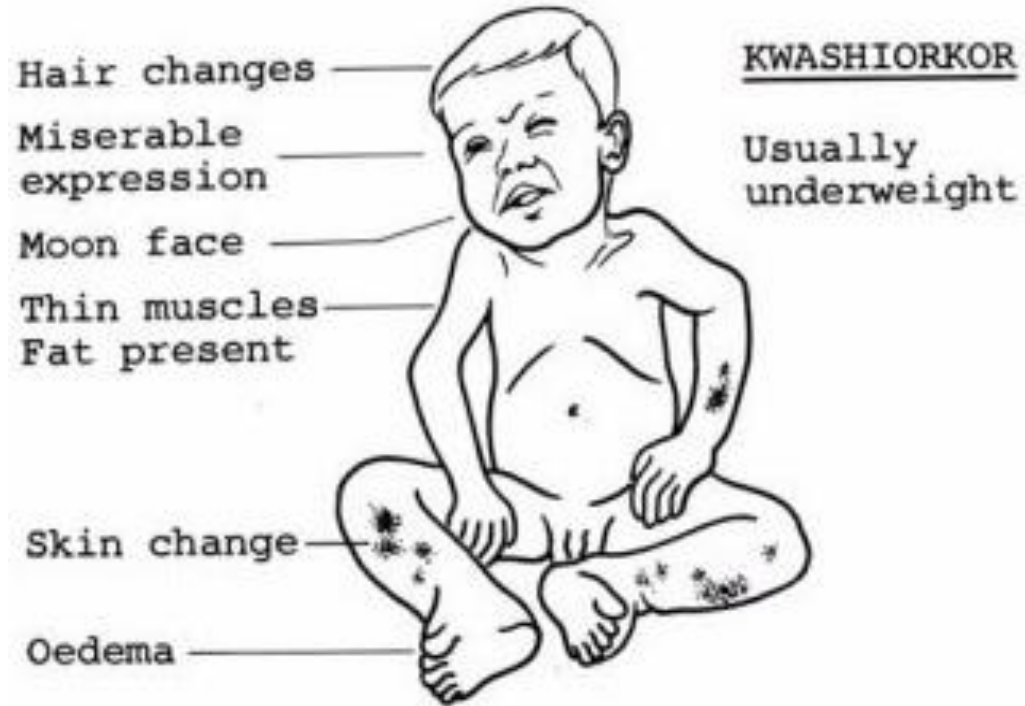
Marasmus versus Kwashiorkor (1)

	Marasmus	Kwashiorkor (edematous malnutrition)
Height and weight	Low weight for height (wasting). Low height for age (stunting), if also chronically malnourished.	Weight for age is variable. Low height for age (stunting) if also chronically malnourished
Head	Appears large relative to body, with staring eyes.	Moon face (rounded prominence of the cheeks), with or without facial edema. May appear plump.
Extremities	Emaciated arms, thighs, and buttocks, with loose folds due to loss of subcutaneous fat.	• Edema: • Mild (1+): involves only the feet • Moderate (2+): involves feet, legs, and upper limbs • Severe (3+): either generalized edema or moderate plus facial edema

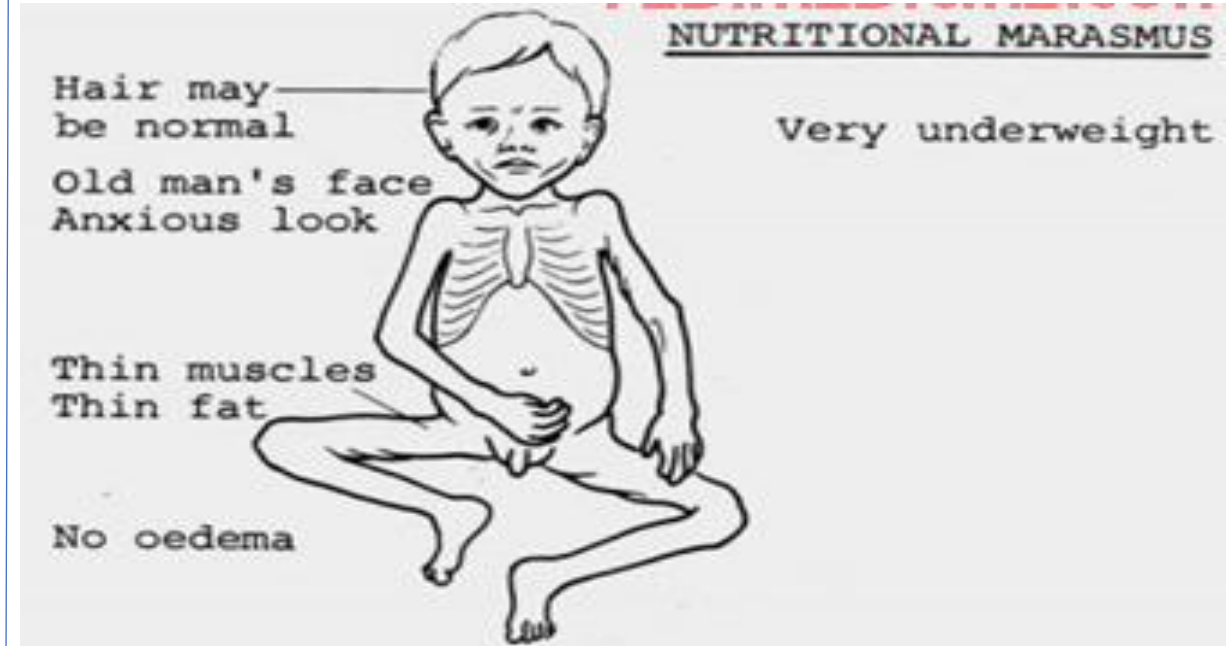
Marasmus versus Kwashiorkor (2)

	Marasmus	Kwashiorkor (edematous malnutrition)
Behavior	Typically irritable, fretful.	Typically apathetic, listless; may be irritable when handled.
Abdomen		Distended, dilated intestinal loops; hepatomegaly.
Skin	Thin, dry skin.	Thin, dry and peeling skin, with areas of hyperkeratosis and hyperpigmentation (dermatosis of kwashiorkor).
Hair	Thin, sparse hair that is easily plucked.	Dry, dull and hypopigmented hair, which falls out or is easily plucked. Hypopigmentation may occur in stripes if malnutrition has been intermittent ("flag sign").

Kwashiorkor versus Marasmus



- Irritable when picked up, apathetic when left alone.
- Appetite is poor.



- Alert and irritable
- Appetite is good

Marasmic-Kwashiorkor

This severe protein-calorie malnutrition is characterized by:

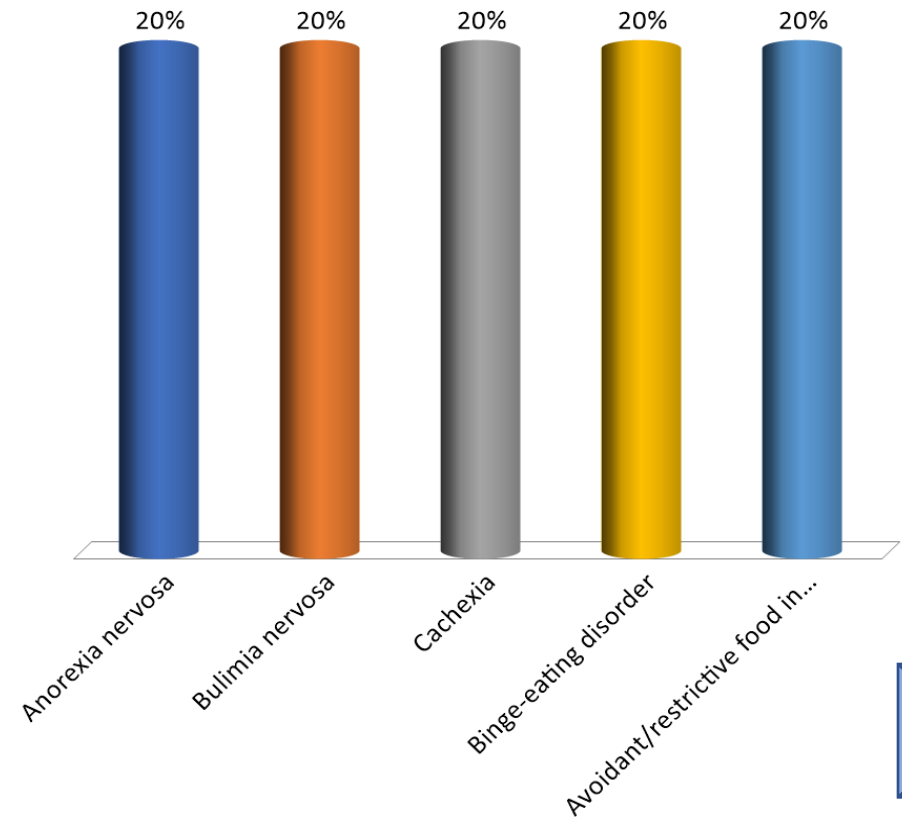
- extreme weight loss (wasting)
- weakness, and
- bilateral pitting edema



Clicker Question

A 17-year-old teenager was brought to the physician by her father because of intermittent vomiting. The patient is a competitive athlete and follows a strict diet and exercise regimen. The father reports that the patient is very self-conscious and worries about her appearance. She wears big clothes most times to hide her body. She has no medical condition. Her BMI is 17.5 kg/m^2 . Physical examination shows dry skin and bradycardia. Which of the following is the most likely diagnosis in this patient?

- ✓ A. Anorexia nervosa
- B. Bulimia nervosa
- C. Cachexia
- D. Binge-eating disorder
- E. Avoidant/restrictive food intake disorder



Anorexia nervosa

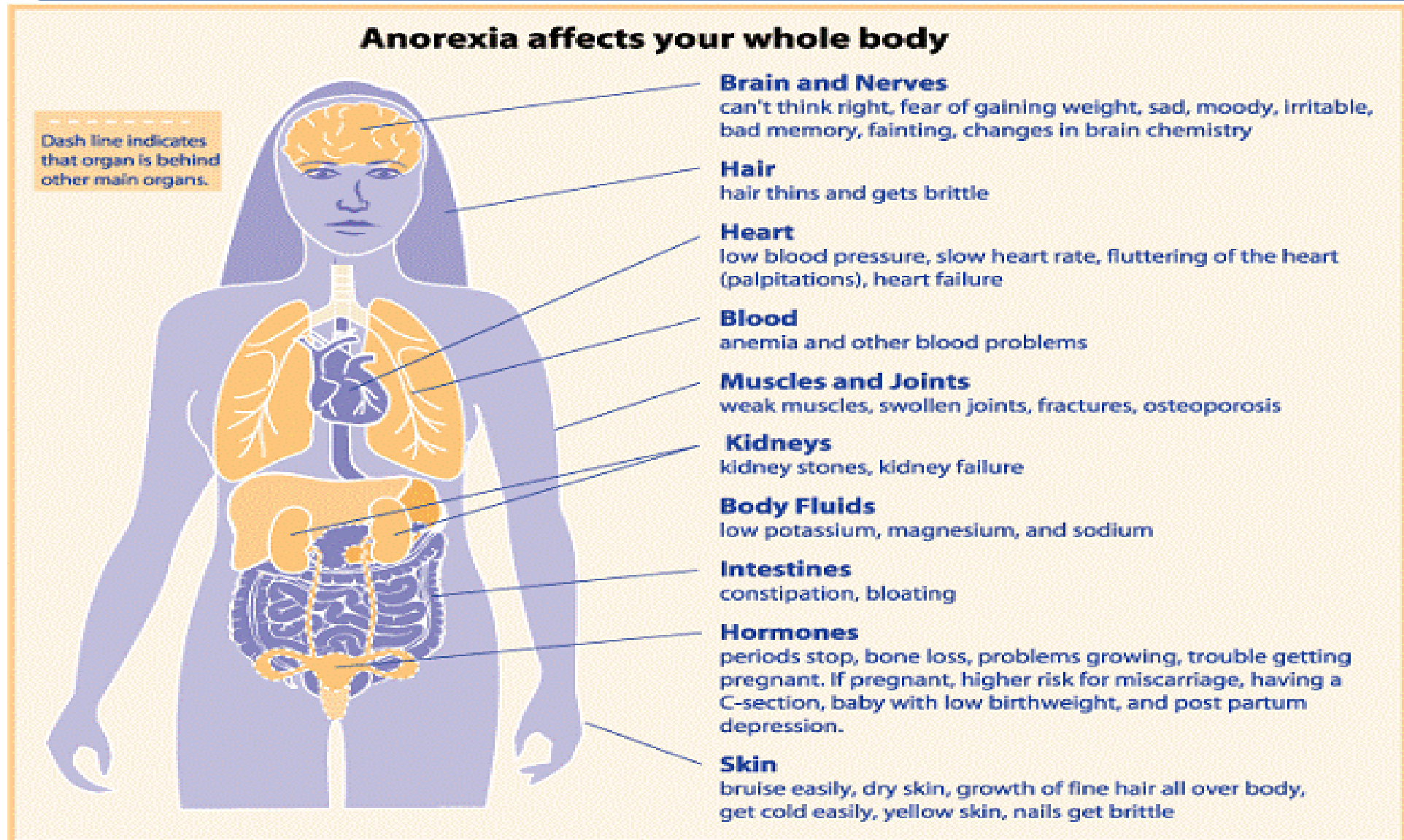
- **Eating Disorder**
- **Diagnosis requires the following criteria:**
 - Restriction of energy intake that leads to a low BMI < 18.5
 - Fear of gaining weight/ persistent behavior to avoid weight gain
 - Distorted perception of body weight and shape
- **Two Subtypes**
 - Restricting food intake
 - Binge eating and Purging



Causes of Anorexia Nervosa

- The cause of anorexia nervosa is unknown. However, it is believed that the following may play a role:
 - Social attitudes toward body appearance
 - Family influences
 - Genetics
 - Brain chemical imbalances
 - Developmental issues

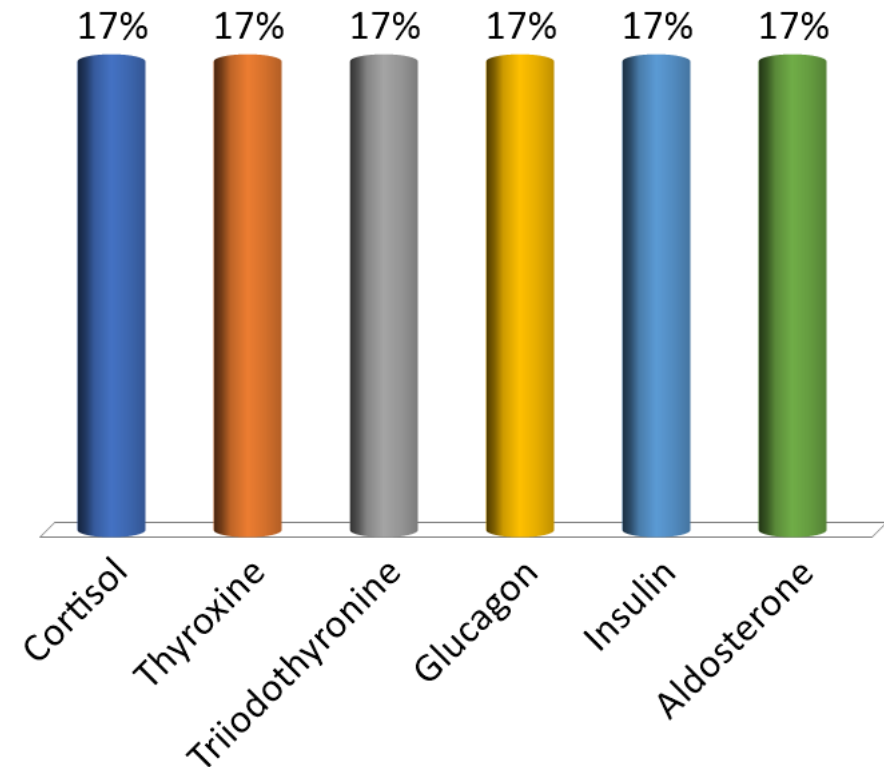
Anorexia affects your whole body



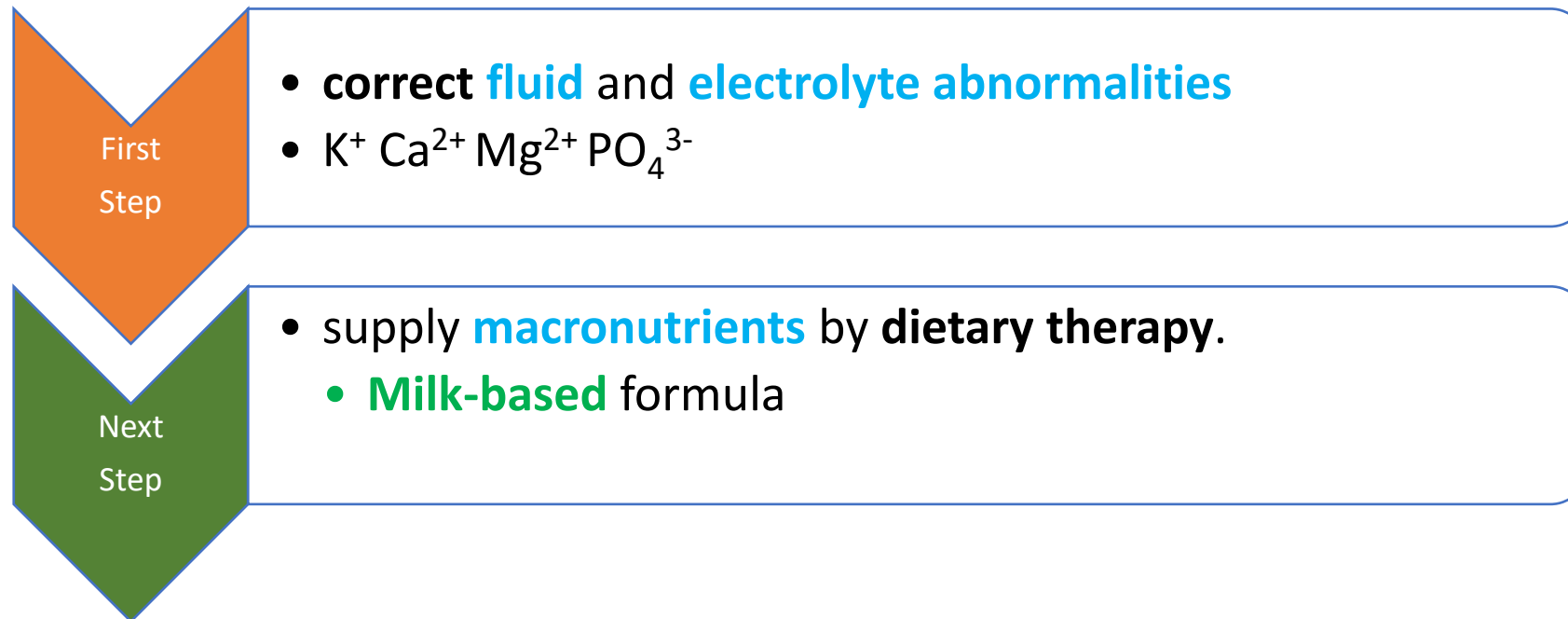
Clicker Question

A 17-year-old girl is admitted to the hospital due to purging behavior, weight loss, and loss of consciousness. Her BMI is 14 kg/m^2 , physical examination shows an emaciated girl with bradycardia. She was started on feeding therapy, and her activities improved. On second day of therapy, she develops severe shortness of breath due to recurrent arrhythmias. Serum chemistry shows hypokalemia, hypophosphatemia, and hypomagnesemia. The elevation of which of the following hormones best explains her second day deterioration?

- A. Cortisol
- B. Thyroxine
- C. Triiodothyronine
- D. Glucagon
- ✓ E. Insulin
- F. Aldosterone

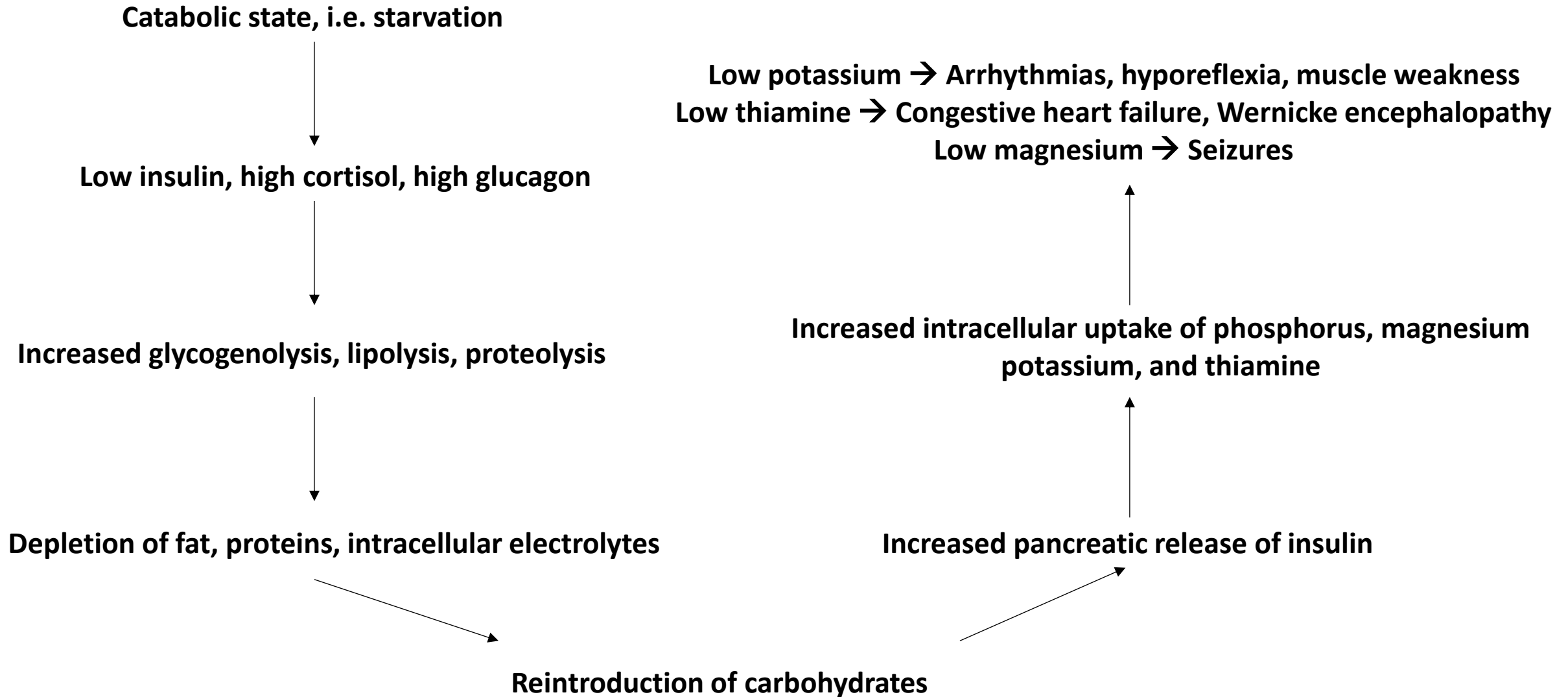


Management of PEM



Step-wise approach is important during correction to prevent a serious complication known as **REFEEDING syndrome**.

How does Refeeding syndrome happen?



Overnutrition (**Obesity**)

- Obesity refers to an excess of fat.
- Obesity is assessed by the BMI. **BMI > 95th percentile for age and sex is considered obese for children between 2-20 yrs.**
- The prevalence of childhood and adolescent obesity has increased rapidly worldwide.
- The increasing prevalence of childhood obesity is related to a **complex combination of socioeconomic, genetic, and biological factors**

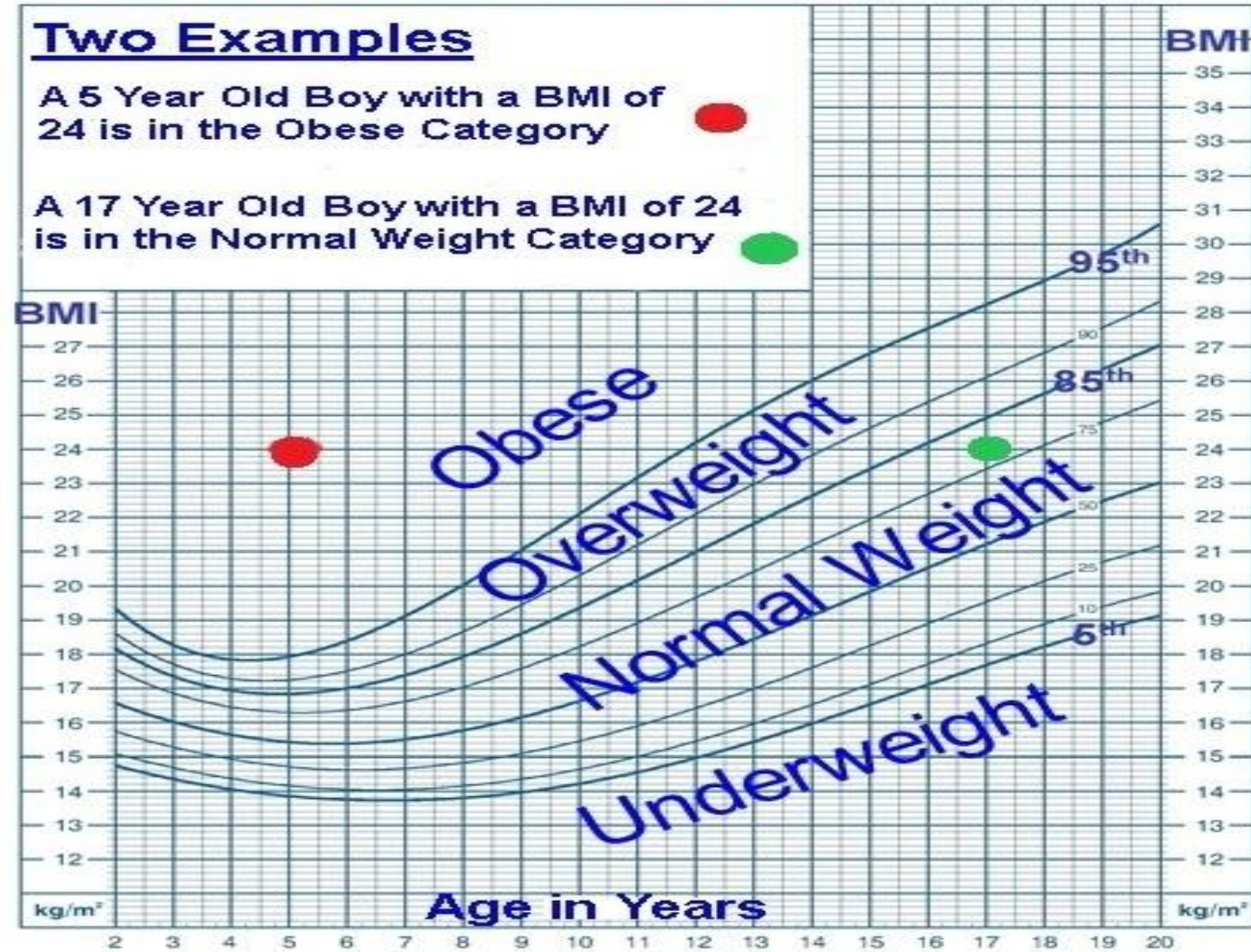
BMI Percentile Chart for Boys Age 2 years to Age 20 years

Two Examples

A 5 Year Old Boy with a BMI of 24 is in the Obese Category



A 17 Year Old Boy with a BMI of 24 is in the Normal Weight Category



Published May 30, 2000 (modified 10/16/00).

SOURCE: Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion (2000).
<http://www.cdc.gov/growthcharts>



SAFER • HEALTHIER • PEOPLE™

Obesogenic Environment

- **Risk factors** in the **home environment**

offer targets for intervention:

- Consumption of **sugar-sweetened beverages**.

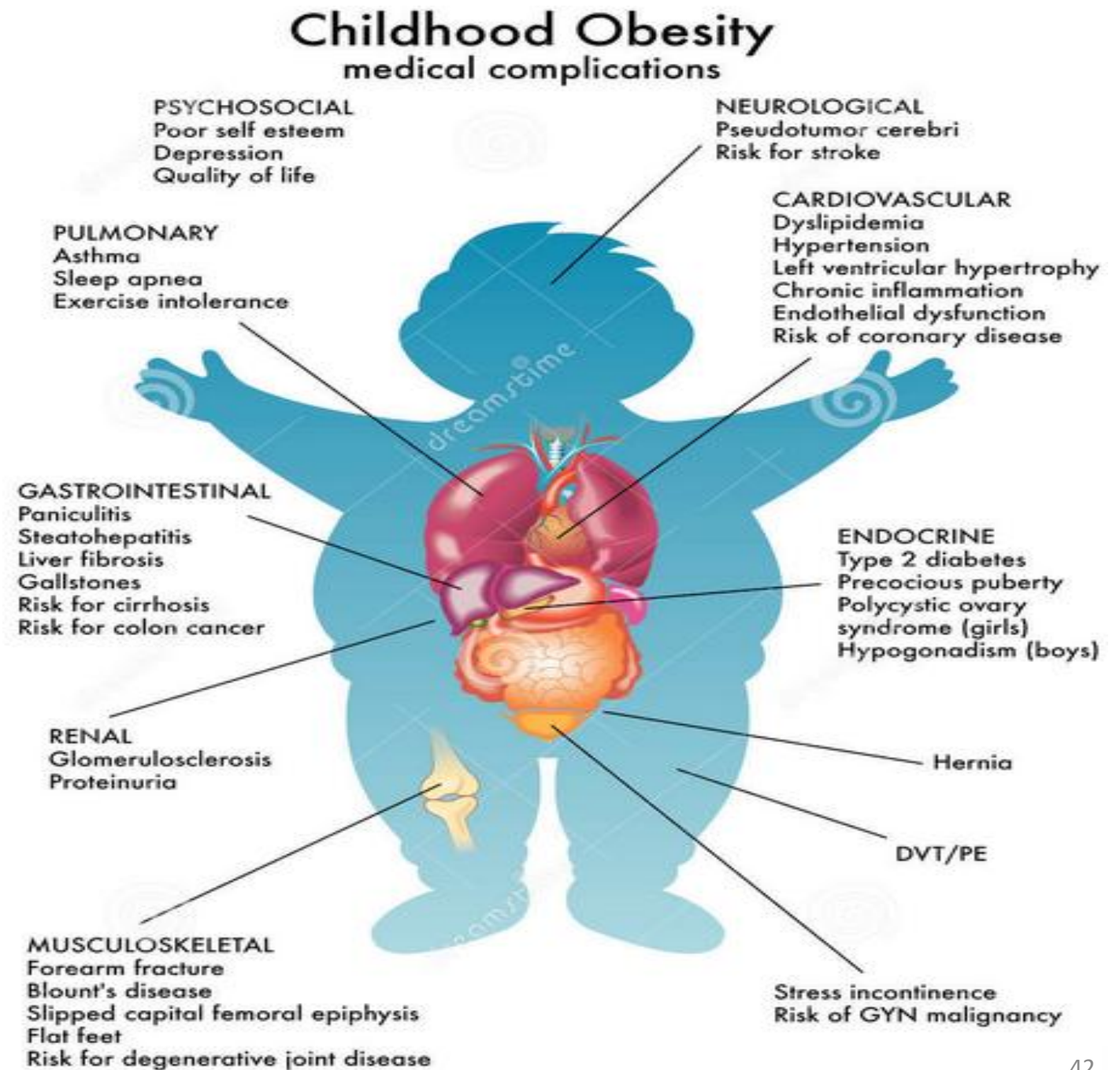
- Lack of family meals, **large portion** sizes and foods prepared outside the home.

- **Television viewing** and **video gaming**.

- Poor sleep, and lack of activity.



Self-study: Medical complications



Considerations in the management of Obesity

Should follow a “**staged approach**”

- **Severity** of overweight and **age** of the child
- Readiness of the **family** to implement changes
- **Weight preferences** of the parents and child and weight management goals
- **Skills** of the health care provider (Pediatrician and Nutritionist)

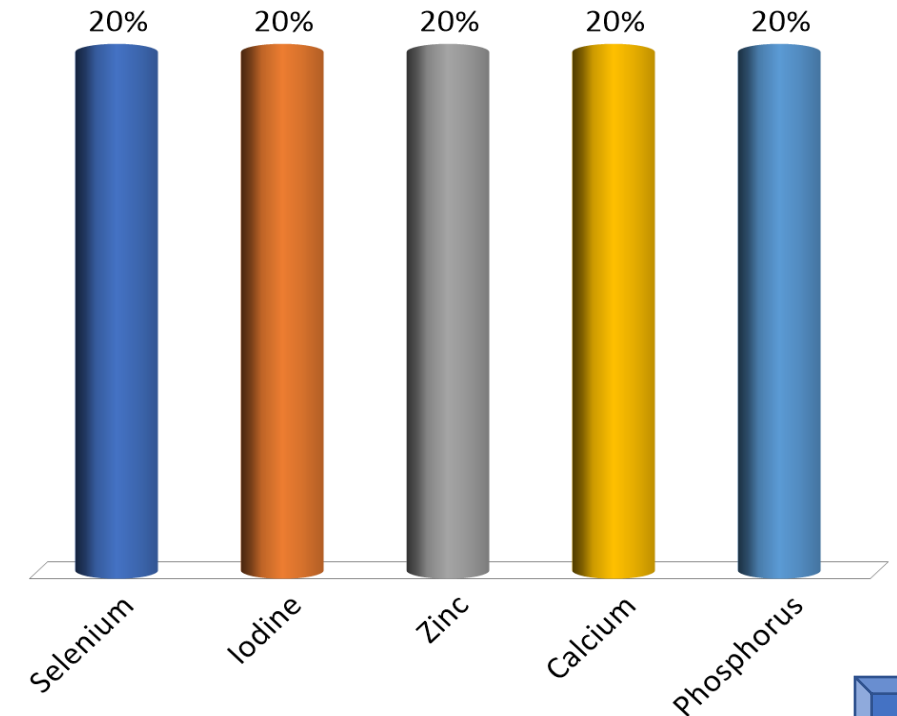
Management of Obesity

- Diet
- Physical activity (play)
- Behavioral Lifestyle modification
- Medical/ drugs
- Surgery

Clicker Question

A 2-year-old boy is brought to the physician by his mother because of weakness and weight loss. She reports frequent non-bloody diarrhea. The boy says that food does not taste the same as before. He has achieved age-appropriate developmental milestones. Physical examination shows patchy alopecia and a pustular, crusting skin rash with scaling and erythema around the mouth and on the extremities. Which of the following is most likely deficient in this patient?

- A. Selenium
- B. Iodine
- ✓ C. Zinc
- D. Calcium
- E. Phosphorus



Macrominerals

Calcium and Phosphorus

Minerals	Function	Deficiency	Clinical Features
Calcium : found in dairy products, legumes, broccoli, green leafy vegetables.	Building block for hydroxyapatite (for osteoid mineralization). Stabilizes resting membrane potential. Second messenger. Coagulation cascade.	Hypovitaminosis D Hypoparathyroidism (e.g., DiGeorge) Hypomagnesemia	Muscle cramps Chvostek & Trousseau signs Tetany Seizures
Phosphorus : meats, eggs, dairy products, grains, legumes, and nuts; high in processed foods and sodas.	Building block for hydroxyapatite, ATP, cell membranes.	Hypovitaminosis D Hyperparathyroidism Refeeding syndrome	Muscle weakness Bone pain Rhabdomyolysis Osteomalacia Respiratory failure Seizures Arrhythmias

Macrominerals

Magnesium and Potassium

Minerals	Function	Deficiency	Clinical Features
Magnesium : Vegetables, cereals, nuts.	Cofactor for many enzymes including adenylate cyclase (needed for PTH activation)	Malnutrition Malabsorption Alcoholism Refeeding syndrome Diarrhea	Hypocalcemia → Neuromuscular excitability, neurologic abnormalities, ECG changes.
Potassium : Nuts, whole grains, meats, fish, beans, fruits and vegetables, especially bananas, orange juice.	Maintain resting membrane potential	Diuretics Glucosuria Hyperaldosteronism Alkalosis	Muscle weakness Rhabdomyolysis Constipation Arrhythmias Respiratory failure

Microminerals

Zinc and Copper

Trace elements	Function	Deficiencies	Features
Zinc : Human milk, meats, shellfish, legumes, nuts, and whole-grain cereals.	Important role in gene transcription and cell division. Important for normal spermatogenesis, fetal growth, and embryonic development.	Malabsorption Total parenteral nutrition Inadequate dietary intake Bowel resection	- Alopecia - Perioral pustular skin rash - Diarrhea - Hypogonadism - Impaired wound healing - Impaired taste - Immune dysfunction
Copper : Human milk, meats, shellfish, legumes, nuts, and whole-grain cereals.	Cytochrome (ETC) Lysyl oxidase (cross linking collagen) Iron absorption Melanin synthesis	Generalized malnutrition, malabsorption, or prolonged diarrhea. Menkes disease.	- Anemia - Delayed wound healing, brittle hair - Skin depigmentation - Muscle weakness - Osteoporosis - Ataxia, peripheral neuropathy

Microminerals

Selenium and Iodine

Trace elements	Function	Deficiencies	Features
Selenium : Seafood, meats, garlic, mushrooms	Cofactor for glutathione peroxidase (protects against free radicals) Required for thyroid hormone synthesis	Malnutrition Total parenteral nutrition	- Skeletal muscle dysfunction → muscle pain and tenderness. - Macrocytosis. - Cardiomyopathy in infants and children. - Thyroid dysfunction.
Iodine : Iodized salt.	Important component of thyroid hormones T3 and T4	Inadequate dietary intake.	- Intellectual disability, deaf mutism, spastic diplegia, and strabismus - Congenital hypothyroidism